Chapter 3 — Chemical Bonding

Exercise Solutions · Full textbook exercise · PECTAA Chemistry 11 · Book pp. 66–69

Q.1 — Multiple Choice Questions (with correct option and reason)

I. Chemical bond formation takes place when

(a) Force of attraction is equal to the force of repulsion

(b) Force of repulsion is greater than force of attraction

(c) Force of attraction overcomes force of repulsion

(d) None of these

Answer: (c) Force of attraction overcomes force of repulsion — A stable bond forms when attractive forces (nucleus–electron) outweigh repulsive forces, lowering the potential energy.

II. An ionic compound A^+B^- is most likely to be formed when

(a) Ionization energy of A is high and electron affinity of B is low.

(b) Ionization energy of A is low and electron affinity of B is high.

(c) Both the ionization energy of A and electron affinity of B are high.

(d) Both the ionization energy of A and electron affinity of B are low.

Answer: (b) Ionization energy of A is low and electron affinity of B is high. — A must lose an electron easily (low I.E.) and B must gain it readily (high/electron-releasing affinity).

III. Which of the following molecules has zero dipole moment?

(a) NH_3

(b) $CHCl_3$

(c) H_2O

(d) BF_3

Answer: (d) BF_3 — BF_3 is trigonal planar and symmetrical, so the three B–F bond dipoles cancel. NH_3, $CHCl_3$ and H_2O are polar.

IV. The numbers of σ and π bonds in the N_2 molecule are:

(a) One σ and one π bonds

(b) One σ and two π bonds

(c) Three σ bonds only

(d) Two σ and one π

Answer: (b) One σ and two π bonds — N≡N consists of one σ bond and two π bonds.

V. Which of the following species has unpaired electrons in antibonding molecular orbitals?

(a) O_2^{2+}

(b) N_2^{2-}

(c) B

(d) F_2

Answer: (b) N_2^{2-} — N_2^{2-} has 12 valence electrons; the two extra electrons occupy the degenerate π*2p orbitals singly (Hund's rule), giving unpaired antibonding electrons.

studysapiens.com

VI. The shape of ICl_5 according to the VSEPR model is:

(a) Tetrahedral
(b) Trigonal planar
(c) Trigonal bipyramidal
(d) T-shape

Answer: No correct option among (a)–(d) — ICl_5 has 5 bond pairs + 1 lone pair (AX_5E) → square pyramidal. Square pyramidal is not among the printed options.

Textbook error (p. 67): the correct shape, square pyramidal, is missing from the options. None of (a)–(d) is correct.

VII. Which of the following molecules has a net dipole moment?

(a) CO_2
(b) CS_2
(c) SO_2
(d) CCl_4

Answer: (c) SO_2 — SO_2 is bent, so its bond dipoles do not cancel. CO_2 and CS_2 are linear; CCl_4 is tetrahedral and symmetrical.

VIII. How many electrons are present in the valence shell of P in PO_4^{3-}?

(a) 8
(b) 10
(c) 12
(d) 14

Answer: (b) 10 — In the expanded-octet Lewis structure P forms five bonds (one P=O and three P–O), giving 10 electrons around P.

IX. How many extra electrons than a normal octet are there in the valence shell of I in ICl_5?

(a) 2
(b) 3
(c) 4
(d) 5

Answer: (c) 4 — I has 5 bond pairs + 1 lone pair = 12 electrons; 12 − 8 = 4 extra electrons.

X. What is the type and shape of $[ICl_4]^-$ according to the VSEPR model?

(a) AB_4, tetrahedral
(b) AB_4, pyramidal
(c) AB_5, trigonal bipyramidal
(d) AB_6, square planar

Answer: (d) AB_6, square planar — $[ICl_4]^-$ has 4 bond pairs + 2 lone pairs (AX_4E_2); the molecular shape is square planar.

Textbook check (p. 67): option (d) gives the correct shape (square planar) but the type should be AB_4E_2 (AX_4E_2), not AB_6.

XI. Which of the following molecules has a central atom with sp^3 hybridization and a tetrahedral electron pair geometry?

(a) BF_3
(b) SO_2
(c) CCl_4
(d) PCl_5

Answer: (c) CCl_4 — CCl_4 has four bond pairs and no lone pair → sp^3, tetrahedral. BF_3/SO_2 are sp^2; PCl_5 is sp^3d.

studysapiens.com

XII. Which of the following species contains a dative bond?

(a) CH_4 **(b)** NaCl

(c) NH_4^+ **(d)** O_2

Answer: (c) NH_4^+ — In NH_4^+ the nitrogen lone pair is donated to H^+, forming a coordinate (dative) covalent bond.

Q.2 — Short Answer Questions

a. Define: (i) Dipole (ii) Bond order (iii) Permanent dipole–permanent dipole force (iv) London dispersion force.

(i) A dipole exists when two equal and opposite charges are separated by a distance; a polar bond has a dipole moment $\mu = q \times d$.

(ii) Bond order = (number of bonding electrons − number of antibonding electrons) / 2. It indicates bond stability and multiplicity.

(iii) Permanent dipole–permanent dipole forces are attractive forces between the positive end of one polar molecule and the negative end of another.

(iv) London dispersion forces are weak, temporary attractive forces arising from instantaneous dipoles caused by momentary uneven electron distribution; they exist in all atoms and molecules.

b. Draw the Lewis (electron dot) structures for: (i) HCN (ii) NCl_3 (iii) CO (iv) O_3 (v) NO_2.

Lewis structures (lone pairs shown as dots; each terminal Cl has three lone pairs). O_3 and NO_2 are resonance hybrids; the double bond is delocalised over the two terminal positions.

Lewis structures

```
(i) HCN        H − C ≡ N :          (one lone pair on N)
(ii) NCl₃            ..
              :Cl − N − Cl:        (N has one lone pair; each Cl has 3 lone pairs)
                   |
                  :Cl:
(iii) CO       : C ≡ O :           (one lone pair on C and one on O)
(iv) O₃        O = O − O  ↔  O − O = O   (bent; central O has one lone pair)
(v) NO₂        O = N − O  ↔  O − N = O   (bent; N has one unpaired electron)
```

c. Xenon trioxide, XeO_3: (i) By counting electron pairs around Xe, explain its shape. (ii) Draw a structure showing partial charges and the direction of the dipole moment.

(i) Xe has 8 valence electrons. It forms three Xe=O bonds and retains one lone pair: 3 bond pairs + 1 lone pair = 4 electron domains (AX_3E). The electron-pair geometry is tetrahedral and the molecular shape is trigonal pyramidal.

(ii) Oxygen is more electronegative, so each O is δ− and Xe is δ+. The three bond dipoles point toward the oxygen atoms; because the molecule is pyramidal they

studysapiens.com

do not cancel, giving a net dipole along the three-fold axis. XeO_3 is polar.

d. Explain the difference between the formation of σ and π bonds in terms of VBT.

A σ bond is formed by head-on (axial) overlap of orbitals along the internuclear axis (s–s, s–p or p–p). Electron density is concentrated between the nuclei; it is strong and allows free rotation. A π bond is formed by side-on overlap of parallel p orbitals above and below the internuclear axis; it is weaker and restricts rotation. A single bond is σ; a double bond is σ + π; a triple bond is σ + 2π.

e. Propanone (acetone): (i) Show how the central carbon forms σ and π bonds through hybridization. (ii) Can propanone make a hydrogen bond with water?

(i) The carbonyl carbon is sp^2 hybridised. Three sp^2 orbitals form three σ bonds (two C–C and one C–O) in a trigonal planar arrangement (~120°). The unhybridised p orbital overlaps side-on with an oxygen p orbital to give the C=O π bond.

(ii) Yes. The carbonyl oxygen has lone pairs and can accept a hydrogen bond from water (O–H···O=C). Propanone cannot donate a hydrogen bond (no O–H), but it mixes with water through this acceptor interaction.

f. Predict the shapes of sulfate ($[SO_4]^{2-}$), borate ($[BH_4]^-$) and tri-iodide ($[I_3]^-$) according to VSEPR.

$[SO_4]^{2-}$: 4 bond pairs, no lone pair on S → tetrahedral. $[BH_4]^-$: 4 bond pairs, no lone pair on B → tetrahedral. $[I_3]^-$: central I has 2 bond pairs + 3 lone pairs (AX_2E_3) → linear.

Naming check (p. 68): $[BH_4]^-$ is the tetrahydridoborate (borohydride) ion, not “borate”; the borate ion is BO_3^{3-}. The shape asked for $[BH_4]^-$ is tetrahedral.

g. Sketch the molecular orbital pictures of π(2p) and π*(2p).

Side-on overlap of two parallel p orbitals gives a bonding π(2p) MO (electron density above and below the axis, no node between nuclei) and an antibonding π*(2p) MO (a nodal plane between the nuclei).

Side-on overlap of p orbitals

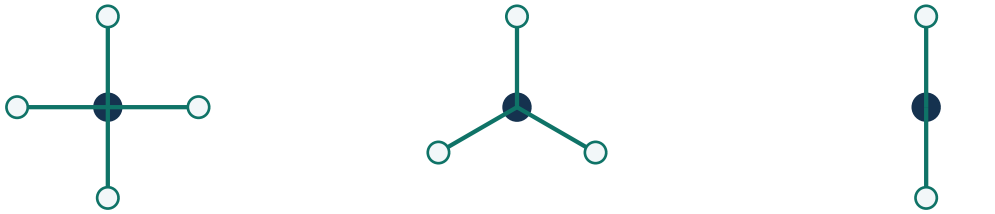


studysapiens.com

h. Sketch the hybrid orbitals and bond formation in PCl_3, $SiCl_4$, and NH_4^+.

PCl_3: P is sp^3 hybridised with one lone pair → trigonal pyramidal. $SiCl_4$: Si is sp^3 with four bond pairs → tetrahedral. NH_4^+: N is sp^3 with four bond pairs (one dative) → tetrahedral.

Hybrid orbital geometries



i. PF_3 and SiF_4: redraw with partial charges and state which is polar and which is non-polar.

In both molecules F is δ− and the central atom is δ+. PF_3 is trigonal pyramidal (one lone pair on P), so the bond dipoles do not cancel → polar. SiF_4 is tetrahedral and symmetrical, so the four bond dipoles cancel → non-polar.

j. Draw the orbital structure of the CO_2 molecule in terms of VBT.

CO_2 is linear, O=C=O. Carbon is sp hybridised: two sp orbitals form two σ bonds with oxygen. The two unhybridised p orbitals on carbon form two π bonds (one with each oxygen) by side-on overlap. Each oxygen also carries two lone pairs.

k. Draw the Lewis structures and tell whether the following ions involve expanded octets: (i) ClO_4^- (ii) ICl_4^- (iii) NH_4^+ (iv) I_3^-.

(i) ClO_4^-: Cl forms four bonds to O (expanded octet, 14 e^- around Cl in the double-bonded form); resonance stabilises the ion. Expanded octet: yes.

(ii) ICl_4^-: central I has 4 bond pairs + 2 lone pairs = 12 e^-; square planar. Expanded octet: yes.

(iii) NH_4^+: N has 4 bond pairs = 8 e^-; tetrahedral. Expanded octet: no.

(iv) I_3^-: central I has 2 bond pairs + 3 lone pairs = 10 e^-; linear. Expanded octet: yes.

Expanded-octet skeletons

```
ClO₄⁻ :   [O–Cl(=O)₃]⁻   (resonance; Cl expanded)
ICl₄⁻ :   square planar, 4 Cl around I + 2 lone pairs
NH₄⁺  :   tetrahedral, 4 H around N (octet)
I₃⁻   :   [I – I – I]⁻  linear, 3 lone pairs on central I
```

studysapiens.com

l. The bond between K and Cl is ionic but that between Si and Cl is polar covalent. Explain why.

K (EN ≈ 0.8) and Cl (EN ≈ 3.0) differ by ~2.2, so an electron is transferred → ionic K^+Cl^-. Si (EN ≈ 1.8) and Cl differ by ~1.2, so electrons are shared unequally → polar covalent Si–Cl.

m. SO_2 is a polar molecule but SO_3 not. Justify.

SO_2 is bent (one lone pair on S), so the two S–O dipoles do not cancel → polar. SO_3 is trigonal planar and symmetrical; the three S–O dipoles cancel → non-polar.

n. Which of O_2^{2+} and O_2^{2-} would be paramagnetic? Give reason in the light of MOT.

Neither is paramagnetic. O_2^{2+} has 10 valence electrons (π* empty) and O_2^{2-} has 14 valence electrons (π* fully filled); in both ions all electrons are paired, so both are diamagnetic.

Textbook check (p. 69): as printed, O_2^{2+} and O_2^{2-} are both diamagnetic. If the intended species were O_2^+ or O_2^-, those would be paramagnetic (one unpaired π* electron).

o. Which of the following bonds would be most polar? (i) C–Cl (ii) Si–F (iii) Se–F.

Si–F is most polar because the electronegativity difference is largest (Si ≈ 1.8, F ≈ 4.0; ΔEN ≈ 2.2). C–Cl (ΔEN ≈ 0.5) and Se–F (ΔEN ≈ 1.6) are less polar.

p. Compare the bond energies of single, double, and triple bonds between the same two atoms (e.g. H–H, O=O, N≡N). Explain the trend in terms of the number of shared electrons.

For the same pair of atoms, bond energy increases and bond length decreases from single to double to triple because more shared electrons increase the attraction between the nuclei and the bonding cloud. Example: H–H ≈ 436, O=O ≈ 498, N≡N ≈ 945 kJ mol^{-1}. (Values for different atom pairs cannot be compared directly.)

studysapiens.com

Descriptive / Long Questions

Q.3 How the bonding in the following molecules can be explained with respect to valence bond theory? (i) Cl_2 (ii) O_2 (iii) N_2 (iv) HF (v) H_2S.

VBT explains covalent bonding by the overlap of half-filled atomic orbitals, with the shared pair localised between the nuclei.

(i) Cl_2: each Cl has a half-filled 3p orbital; head-on 3p–3p overlap forms a σ bond (Cl–Cl).

(ii) O_2: each O has two half-filled 2p orbitals; one p–p head-on overlap gives a σ bond and one side-on overlap gives a π bond → O=O double bond.

(iii) N_2: each N has three half-filled 2p orbitals; one head-on overlap gives a σ bond and two side-on overlaps give two π bonds → N≡N triple bond.

(iv) HF: the half-filled 1s of H overlaps with the half-filled 2p of F to form a σ bond; the bond is polar because F is more electronegative.

(v) H_2S: S uses two half-filled 3p orbitals to overlap with the 1s orbitals of two H atoms, forming two σ bonds; the molecule is bent.

Q.4 What are the postulates of VSEPR model? Discuss the structures of: (i) CH_4 (ii) NH_3 (iii) H_3O^+ (iv) PCl_5 (v) SO_2 (vi) SF_6.

Postulates: electron pairs around the central atom arrange themselves as far apart as possible to minimise repulsion; repulsion order is lone pair–lone pair > lone pair–bond pair > bond pair–bond pair; multiple bonds are treated as one electron domain; the molecular shape is defined by the positions of atoms only.

Species	Electron domains	Lone pairs	Shape
CH_4	4	0	Tetrahedral (109.5°)
NH_3	4	1	Trigonal pyramidal (~107°)
H_3O^+	4	1	Trigonal pyramidal
PCl_5	5	0	Trigonal bipyramidal
SO_2	3	1	Bent (~119°)
SF_6	6	0	Octahedral (90°)

Q.5 Explain the orbital hybridization for CH_4, NH_3, BF_3, and $BeCl_2$.

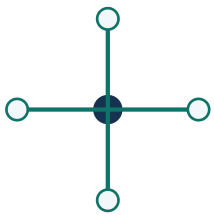
CH_4: carbon promotes to $2s^1\,2p^3$ and forms four sp^3 hybrids → tetrahedral, 109.5°.

NH_3: nitrogen is sp^3 hybridised; three hybrids form N–H bonds and one holds a lone pair → trigonal pyramidal, ~107°.

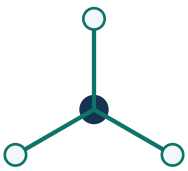
BF_3: boron forms three sp^2 hybrids → trigonal planar, 120°; one empty p orbital remains.

$BeCl_2$: beryllium forms two sp hybrids → linear, 180°.

Hybrid orbital geometries



sp^3 (109.5°)



sp^2 (120°)



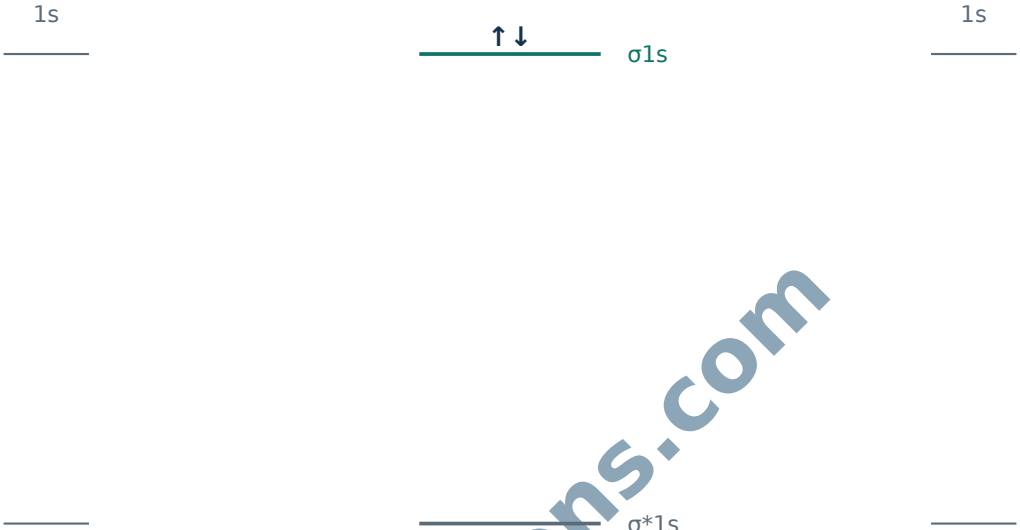
sp (180°)

studysapiens.com

Q.6 Draw the molecular orbital diagrams of the following molecules. Calculate their bond orders. (i) H_2 (ii) He_2 (iii) N_2 (iv) O_2.

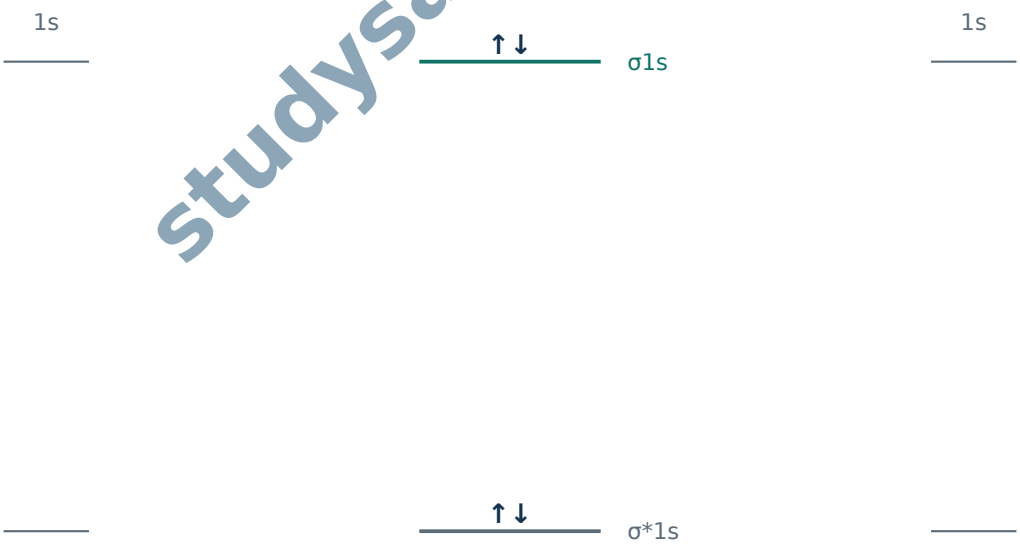
Bond order = (bonding electrons − antibonding electrons) / 2. H_2: (2−0)/2 = 1 (stable). He_2: (2−2)/2 = 0 (does not exist). N_2: (8−2)/2 = 3 (triple bond, diamagnetic). O_2: (8−4)/2 = 2 (double bond, paramagnetic because of two unpaired π* electrons).

MO diagram (valence shell) — H2



Bond order = (2 − 0) / 2 = 1

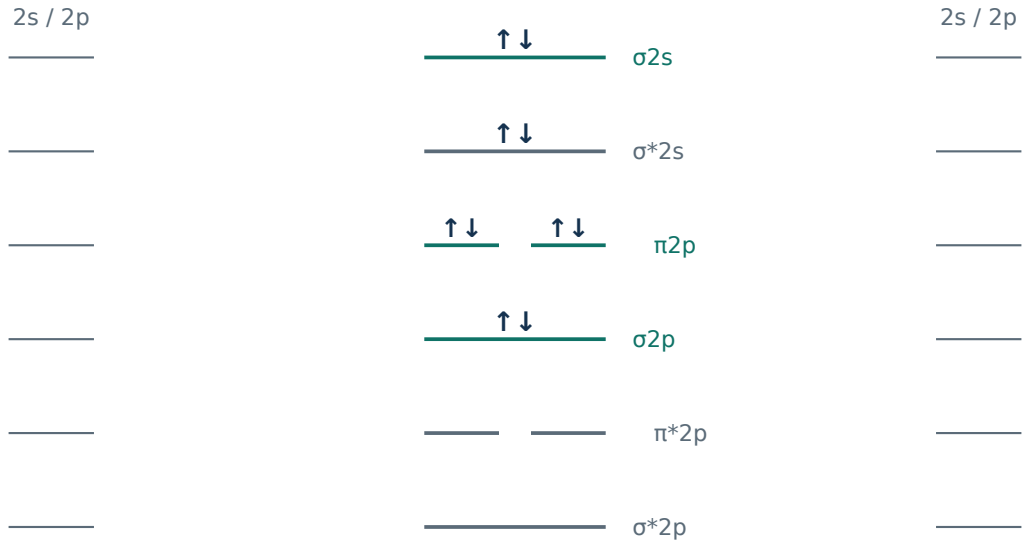
MO diagram (valence shell) — He2



Bond order = (2 − 2) / 2 = 0

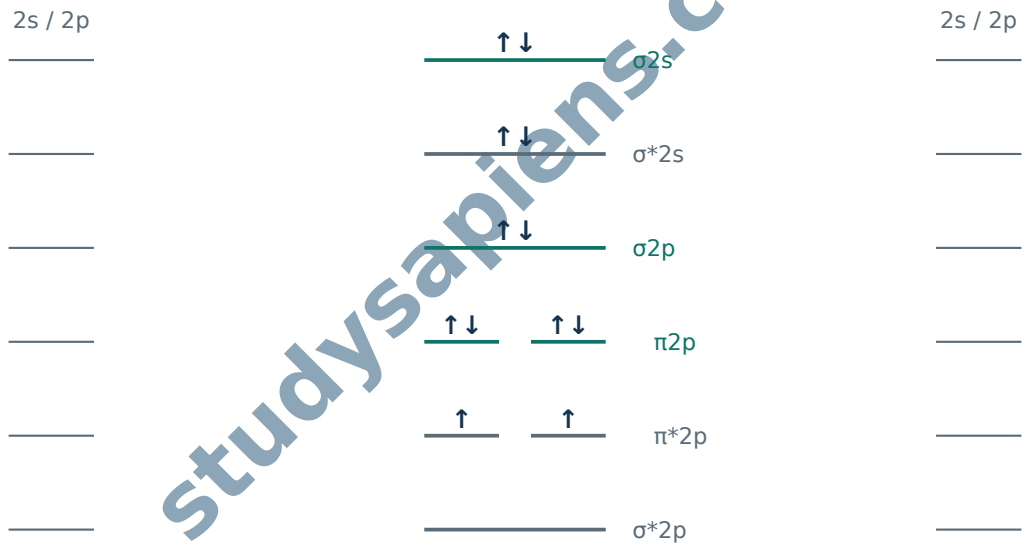
studysapiens.com

MO diagram (valence shell) — N2



Bond order = (8 − 2) / 2 = 3

MO diagram (valence shell) — O2



Bond order = (8 − 4) / 2 = 2

studysapiens.com

Q.7 Discuss the formation of F_2 molecule in the light of Lewis concept, VBT, and MOT.

Lewis: each F has 7 valence electrons; they share one pair to complete octets, giving F–F with three lone pairs on each F.

VBT: a half-filled 2p orbital on each F overlaps head-on to form a σ(2p–2p) bond.

MOT: F_2 has 14 valence electrons: $\sigma 2s^2\ \sigma^*2s^2\ \sigma 2p^2\ \pi 2p^4\ \pi^*2p^4$. Bond order = (8−6)/2 = 1. All electrons are paired, so F_2 is diamagnetic.

MO diagram (valence shell) — F2



Bond order = (8 − 6) / 2 = 1

Quick Check Solutions (in-chapter practice boxes)

Quick Check 3.1 (book p. 46)

a. Draw the Lewis structures of N_2 and CS_2.

N_2: :N≡N: (one lone pair on each N). CS_2: S=C=S with two lone pairs on each S (linear).

b. How many electrons in the valence shell of B in BF_3? Can it accept a lone pair?

B has only 6 electrons (three B–F bonds). It has an empty orbital and can accept a lone pair, acting as a Lewis acid.

c. Show the formation of a dative bond between NH_3 and BF_3.

NH_3 donates its lone pair into the empty orbital of B: $H_3N{\rightarrow}BF_3$. The N→B bond is a coordinate (dative) covalent bond.

Quick Check 3.2 (book p. 47)

a. Can period-2 elements have expanded octets? Explain.

No. Period-2 elements have only 2s and 2p orbitals (maximum 8 electrons) and no available d orbitals, so they cannot expand their octet.

c. Predict and explain the expanded octets in ClO_3^- and PO_4^{3-}.

ClO_3^-: Cl can accommodate more than 8 electrons (expanded octet, ~10 e^- in the double-bonded form). PO_4^{3-}: P forms five bonds in the expanded-octet structure (10 e^- around P). Both use available d orbitals to expand.

studysapiens.com

Quick Check 3.3 (book p. 47)

a. Predict bond type using electronegativity: (i) HF (ii) KI (iii) CaF_2 (iv) ICl (v) Br_2.

(i) HF: polar covalent. (ii) KI: ionic. (iii) CaF_2: ionic. (iv) ICl: polar covalent. (v) Br_2: non-polar covalent.

Quick Check 3.4 (book p. 49)

a. Why CO has a dipole moment but CO_2 does not.

CO is diatomic with unequal sharing, so it has a net dipole. CO_2 is linear and symmetrical (O=C=O), so the two bond dipoles cancel.

b. Are individual C–Cl bonds in CCl_4 polar? What about the overall molecule?

Yes, each C–Cl bond is polar (Cl is more electronegative). But CCl_4 is tetrahedral and symmetrical, so the four bond dipoles cancel and the molecule is non-polar.

c. Polar or non-polar: HF, CH_2Cl_2, O_2, H_2S.

HF: polar. CH_2Cl_2: polar (asymmetric tetrahedral). O_2: non-polar. H_2S: polar (bent).

Quick Check 3.5 (book p. 51)

a. HI is a stronger acid and robust reducing agent, whereas HF is a weaker acid. Explain.

The H–I bond is long and weak, so HI ionizes readily (strong acid) and I^- loses electrons easily (strong reducing agent). The H–F bond is very strong, so HF ionizes poorly (weak acid).

b. Acetylene (HC≡CH) is more stable than ethene ($H_2C=CH_2$). Explain.

The statement needs care: acetylene has a stronger C≡C bond and stronger sp C–H bonds (more s-character), but it is thermodynamically less stable (higher enthalpy) than ethene. If “stable” refers to bond strength, the sp-hybridised bonds are shorter and stronger.

Quick Check 3.6 (book p. 53)

a. Draw the orbital structures of H_2O and N_2.

H_2O: O is sp^3 hybridised; two sp^3 orbitals form O–H σ bonds and two hold lone pairs → bent. N_2: each N uses p orbitals; one head-on p–p overlap gives a σ bond and two side-on overlaps give two π bonds (N≡N).

b. Draw the orbital overlap for F_2 and HF.

F_2: head-on overlap of a half-filled 2p orbital on each F gives a σ(2p–2p) bond. HF: overlap of H 1s with a half-filled F 2p orbital gives a σ(s–p) bond.

studysapiens.com

Quick Check 3.7 (book p. 60)

a. Number of lone pairs on the central atom in H_2O, ICl_3, I_3^-.

H_2O: 2 lone pairs on O. ICl_3: 2 lone pairs on I. I_3^-: 3 lone pairs on the central I.

b. Bond pairs and lone pairs in SiH_4, H_2Se, PH_3.

SiH_4: 4 bond pairs, 0 lone pairs. H_2Se: 2 bond pairs, 2 lone pairs. PH_3: 3 bond pairs, 1 lone pair.

c. Shapes and angles in SiH_4, H_2Se, PH_3.

SiH_4: tetrahedral, 109.5°. H_2Se: bent, ~90°. PH_3: trigonal pyramidal, ~93°.

d. How the bond angle in H_2S differs from H_2O.

H_2S has a smaller bond angle (~92°) than H_2O (104.5°) because S is larger and less electronegative; the bond pairs are farther from the central atom and repel less, allowing a smaller angle.

Quick Check 3.8 (book p. 62)

Draw Lewis structures for IF_3 and IF_5 and predict geometry by VSEPR.

IF_3: central I has 3 bond pairs + 2 lone pairs (AX_3E_2) → T-shaped. IF_5: central I has 5 bond pairs + 1 lone pair (AX_5E) → square pyramidal.

Quick Check 3.5(b): the claim that acetylene is “more stable” than ethene is misleading; acetylene has stronger bonds but higher enthalpy (less thermodynamically stable).

Textbook checks flagged in this chapter

• p. 67, MCQ VI: the correct shape of ICl_5 (square pyramidal) is missing from the options.

• p. 67, MCQ X: $[ICl_4]^-$ is square planar, but its VSEPR type is AB_4E_2, not the printed AB_6.

• p. 68, Q.2(f): $[BH_4]^-$ is tetrahydridoborate (borohydride), not borate.

• p. 69, Q.2(n): as printed, both O_2^{2+} and O_2^{2-} are diamagnetic; the question appears to intend O_2^+/O_2^-.

studysapiens.com